

Topological Pumping in a Floquet-Bloch Band

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Constructing new topological materials is of vital interest for the development of robust quantum applications. However, engineering such materials often causes technological overhead, such as large magnetic fields, spin-orbit coupling, or dynamical superlattice potentials. Simplifying the experimental requirements has been addressed on a conceptual level—by proposing to combine simple lattice structures with Floquet engineering—but there has been no experimental implementation. Here, we demonstrate topological pumping in a Floquet-Bloch band using a plain sinusoidal lattice potential and two-tone driving with frequencies ω and 2ω . We adiabatically prepare a near-insulating Floquet band of ultracold fermions via a frequency chirp, which avoids gap closings en route from trivial to topological bands. Subsequently, we induce topological pumping by slowly cycling the amplitude and the phase of the 2ω drive. Our system is well described by an effective Shockley model, establishing a novel paradigm to engineer topological matter from simple underlying lattice geometries. This approach could enable the application of quantized pumping in metrology, following recent experimental advances on two-frequency driving in real materials.

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The quantization of charge transport in a Thouless pump shares its topological origin with the quantum Hall effect, resulting from the Chern invariant defined on a torus geometry [1]. In the case of the topological Thouless pump, the torus is spanned by one spatial and one temporal dimension, requiring a one-dimensional insulator and a cyclic parameter change, respectively. The topology of the insulator is determined by the Zak phase, which is the Berry phase for energy bands [2]. An adiabatic change of the Zak phase by 2π induces charge transport of each particle by 1 unit cell. Equivalently, this process can be understood as the shift of Wannier centers, i.e., the electric polarization [3], from one lattice site to the next.

So far, topological pumps have been realized in photonics [4–7] and ultracold atoms [8–11], while generalized pumps have also been studied using individual two-level systems [12,13]. Most implementations of topological pumps rely on dynamical superlattices to realize an effective Rice-Mele model [5,8–10]. However, the Rice-Mele pump is not suitable for real materials where the underlying lattice structure is essentially fixed. In this Letter, we demonstrate a topological pump in a simple sinusoidal potential, enabled by Floquet engineering [14–18]. Resonant driving leads to an effective Shockley model [3,19–21], serving as a fundamentally different paradigm for topological pumping. In contrast to the Rice-Mele model, which has two spatially separated lattice sites per unit cell, the two orbitals in the Shockley model are on the same lattice site. This allows the unambiguous realization of a topological transition in one dimension [21] and enables the appearance of fractionalized edge modes [22].

We use ultracold fermionic spin-polarized ^{40}K atoms that are initially prepared in the s band of a simple optical standing wave (see Supplemental Material [23]). The optical lattice is a sinusoidal potential $V(x) = V_0 \cos^2(\pi x/a)$, with lattice spacing $a = 532$ nm, depth $V_0 = 4E_r$, and recoil energy $E_r = \hbar^2/8ma^2 = \hbar \times 4.41$ kHz, where $\hbar$ is Planck's constant and m is the mass of ^{40}K . The lattice is reflected off a piezoactuated mirror, which is driven by an arbitrary waveform generator, resulting in a force $F(\tau) = (\hbar\omega/a)[K_1 \cos(\omega\tau) + 2K_2 \cos(2\omega\tau + \varphi)]$. This force on charge-neutral atoms is analogous to an effective ac electric field for electrons in a solid. The two-tone driving waveform [24–31] is parametrized by the dimensionless amplitudes K_1 and K_2 of the ω and 2ω drives, respectively. Each harmonic can hybridize the s and p bands [Figs. 1(a)–1(c)] [28,30,32,33] leading to an effective two-band model with band inversion points (Shockley model [3,19–21] or generalized Creutz ladder [34–36]). The relative phase φ defines whether time-reversal symmetry is preserved ($\varphi = 0, \pi$) or broken ($\varphi \neq 0, \pi$). In the case of broken time-reversal symmetry, the resulting Floquet-Bloch bands are asymmetric, enabling individual gap closings in the quasienergy spectrum [32]. For $\varphi = \pm\pi/2$, the system fulfills a time-glide symmetry, which is an intrinsically dynamical, nonsym-morphic symmetry [32,37–39] and at a certain critical amplitude K_2 a single band gap closes. This is a topological transition point characterized by a Zak phase winding of 2π [Fig. 1(d)], as pointed out by Kang and Shin [36]. If a single Floquet-Bloch band is uniformly filled, a topological pump will result from the cyclic variation of the driving

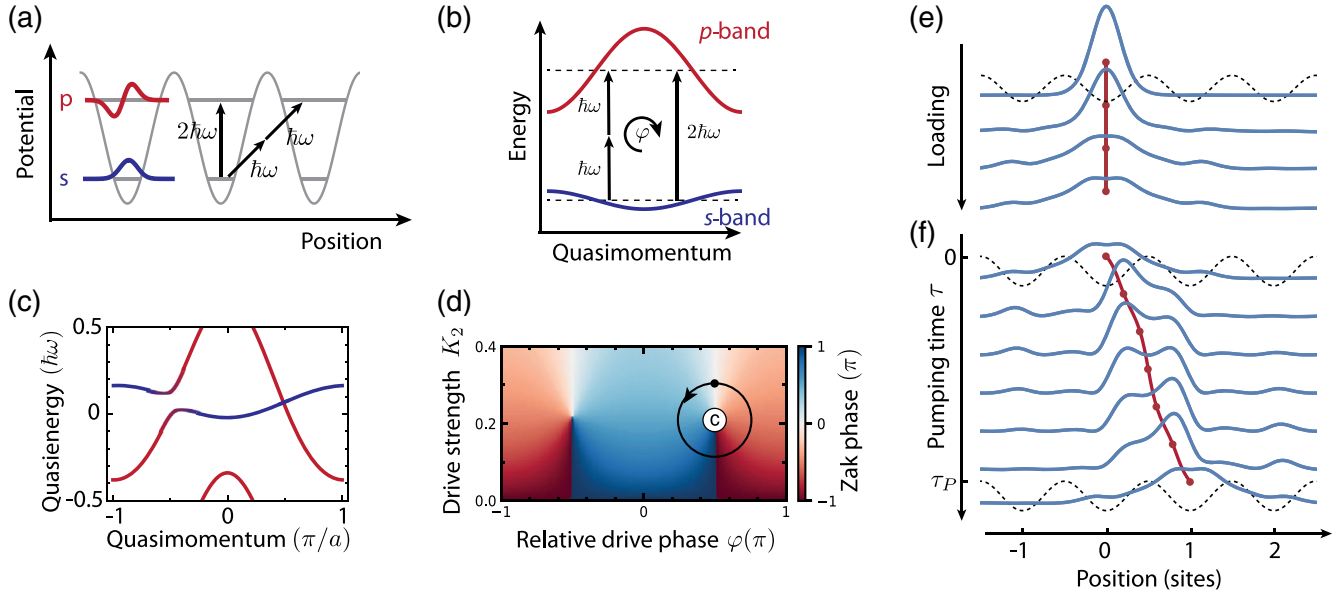


FIG. 1. Floquet-Bloch bands and topological pump in a resonantly shaken optical lattice. (a) s and p orbitals of a sinusoidal potential are coupled on site and between neighboring sites via one-photon (2ω) or two-photon ($\omega + \omega$) resonances, respectively. (b) One- and two-photon resonances interfere constructively or destructively, depending on the relative phase φ and quasimomentum q . (c) Floquet-Bloch bands for critical parameters ($\omega/2\pi$) = 6.5 kHz, $K_1 = 0.8$, $K_2 = 0.22$, and $\varphi = +\pi/2$. The bands are asymmetric due to broken time-reversal symmetry and the band gap closes around quasimomentum $q = +(\pi/2a)$ while it remains open at $q = -(\pi/2a)$. (d) Zak phase of the “lower” Floquet-Bloch band with topological transition points at $K_2 = 0.22$ and $\varphi = \pm\pi/2$. A counterclockwise orbit starting at $\varphi = +\pi/2$ (black dot) around the critical point marked as “c” (band structure in c) winds the Zak phase by $+2\pi$. (e) During Floquet state preparation (vertical axis, “loading”), the initial s -band Wannier state (blue line, top) obtains on-site and neighboring-site p -band contributions. (f) Topological pumping along the counterclockwise orbit shown in (d) (vertical axis, “Pumping time τ ”) shifts the Wannier center (red line) by one lattice site per pump period τ_P .

parameters around the critical point [Figs. 1(e) and 1(f)]. Contrary to previous works on directed group velocities due to time-reversal symmetry breaking [28,40], the Floquet-engineered Thouless pump has a geometric origin [1].

A key aspect of our Letter is the ability to connect the initially occupied s band to the Floquet band without gap closing processes. Here, we follow a three-step protocol that is outlined in Fig. 2, motivated by Ref. [36]. First, the 2ω drive is switched on at a frequency red-detuned to the $s - p$ band gap. The dispersion of the occupied Floquet-Bloch band is similar to the dispersion of the static s band. Second, the 2ω frequency is continuously increased, ending at its target value resonant to the $s - p$ band gap. During this stage the bands hybridize and the band gaps move toward the BZ center, reaching approximately $q = \pm(\pi/2a)$ at the end of the ramp. Third, the amplitude K_1 is linearly increased from zero to its final value, which creates an asymmetry between the two band gaps if time-reversal symmetry is broken. After the preparation, the minimal band gap ($\hbar \times 0.26$ kHz) is located at approximately $q = +(\pi/2a)$ for $\varphi = +\pi/2$, as indicated by the orange arrows in Fig. 2.

The topological pump relies on adiabatically connecting the static system to a single Floquet band. The target band has p -band character at the BZ edges, shown in red in Figs. 2(b), 2(c), and 2(e), whereas it has s -band character at

the BZ center, indicated in blue. Thus, we can experimentally determine the adiabatic timescale for Floquet loading by counting p -band atoms in a band-mapping sequence and varying the frequency ramp duration followed by a hold time (see Supplemental Material [23]). The experimental data exhibits an increase of p -band population as function of ramp time, saturating at 5 ms to a value of around 35% [orange triangles in Fig. 2(f)]. Exemplary band-mapping images show p -band atoms in the outer parts of the BZ appearing at the expense of s -band atoms from the inner BZ [Figs. 2(g) and 2(h)]. The measurements of p -band population suggest an adiabatic timescale of around 5 ms, which is consistent with an estimate from the Landau-Zener formula given by the minimal gap 0.72 kHz to the “excited” Floquet band during the frequency ramp (see Supplemental Material [23]). We confirm this by reversing the loading process and transferring essentially all atoms back to the s band for ramp times longer than 5 ms (green diamonds, *In-Out*). While the loading protocol achieves adiabaticity with respect to band population, it also leads to redistribution of quasimomentum states [Fig. 2(i)]. The quasimomentum dynamics is attributed to the expansion of the atomic cloud since we switch off the confining potential along the lattice direction prior to shaking (see Supplemental Material [23]). Similar measurements for the K_1 ramp produce an adiabatic timescale of 2 ms

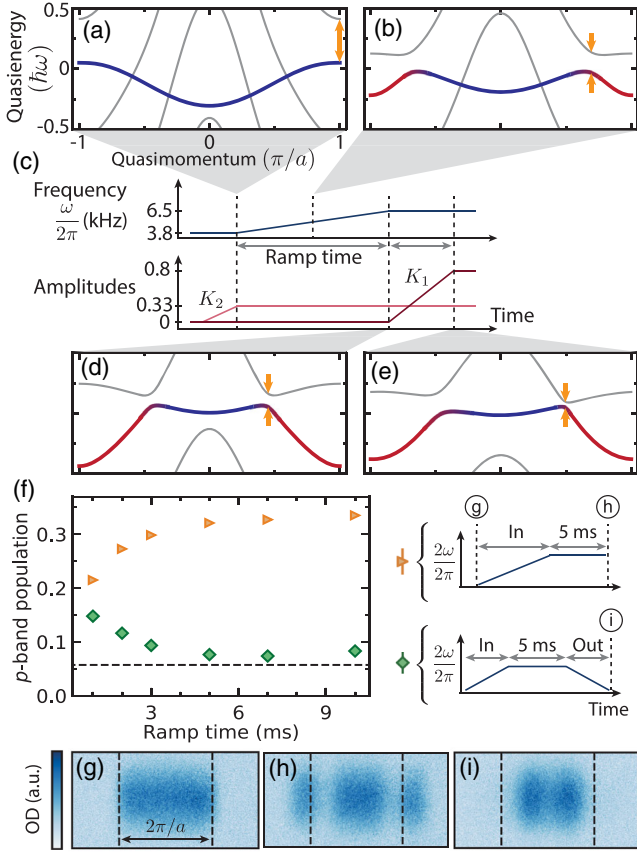


FIG. 2. Adiabatic Floquet state preparation. (a)–(e) Floquet-Bloch quasienergy bands and parameter ramps during the loading sequence. Occupied s and p bands are colored in blue and red, respectively, while empty bands are displayed in gray. (a) Initially, the s band (blue) is separated by a large gap from the p band (orange arrow at the Brillouin zone (BZ) edge). (b), (d) During the frequency ramp (c) the minimal band gap (orange arrows) “slides in” from the BZ edge and the outer portion of the Floquet band adiabatically changes from s to p type (red). (e) The target Floquet state is asymmetric in quasimomentum at relative phase $\varphi = \pi/2$ due to broken time-reversal symmetry. During the loading sequence there is no gap closing. (f) Experimental frequency ramp optimization, showing p -band population versus ramp time for *In* (orange triangles) and *In-Out* (green diamonds) sequences. Each data point is the mean and standard error of the mean (SEM) of three repetitions; the error bar is smaller than the marker size. The horizontal dashed line indicates the static situation, in which s -band population is spuriously counted in the second BZ. (g)–(i) Raw optical density (OD) images after band mapping in the static lattice (g), after a 5 ms *In* sequence (h), and after a 5 ms *In-Out* sequence (i), respectively. The lattice potential is oriented horizontally, corresponding to the laboratory x direction (see Supplemental Material [23]). Vertical dashed lines mark the BZ edges.

(see Supplemental Material [23]). In the following, we use ramp times of 5 ms and 2 ms for the 2ω and K_1 ramps, respectively, during Floquet state preparation. The optimized loading sequence initializes the Floquet state in the (K_2, φ) plane at the points with $K_2 = 0.33$ and

$\varphi = 0, \pm\pi/2$, which are used as a starting point for all subsequent pumping orbits.

We demonstrate pumping by cycling the drive parameters around topological transition points in the (K_2, φ) plane and recording the *in situ* center of mass (c.m.) position of the atomic cloud (see Supplemental Material [23]). The lattice loading and the state preparation sequence can lead to a deviation from a homogeneous distribution of quasimomentum states in the Floquet-Bloch band [Fig. 2(i)]. Thus, group velocity effects can cause (non-topological) directional motion due to the asymmetry in the band structure, similar to Refs. [28,40]. To overcome this, the c.m. position shift is averaged over paired orbits designed to cancel group velocity effects while inducing the same Zak phase winding, as illustrated in Fig. 3. For example, the blue clockwise and counterclockwise orbits around $\varphi = +\pi/2$ and $-\pi/2$, respectively, each change the Zak phase by -2π but exhibit opposite band asymmetry. In the experiment, we measure an average c.m. shift for this pair of orbits consistent with topological pumping [blue diamonds in Fig. 3(a)]. For the red orbits, which wind the Zak phase by $+2\pi$, we observe the opposite c.m. shift (red squares), whereas the gray orbits, which give no net Zak phase winding, lead to no displacement (gray circles). These results are in agreement with numerical simulations (blue, gray, and red solid lines), based on solving the time-dependent Schrödinger equation with an effective two-band Hamiltonian, assuming a homogeneous distribution of quasimomentum states (see Supplemental Material [23]). Nonadiabatic effects and deviations from a homogeneous population of the band result in a below unit efficiency. In contrast to previous realizations of topological pumps in superlattices [8,9], the filled band remains dispersive throughout the pumping process. Therefore, individual orbits can lead to significant motion of the atoms (approximately five lattice sites, Fig. 3(a) inset), as expected for an imperfect band insulator with high tunnelling rates.

To highlight the importance of the loading protocol, we complement our previous measurements with identical parameter cycles but without adiabatic state preparation, i.e., with a sudden switch-on of the drives at the beginning of the each orbit [Fig. 3(b)]. In such an experiment we observe no average c.m. shift for any pair of orbits (blue, red, or gray). Numerical simulations assuming an initial s -band insulator agree not only with the averaged shifts (blue, gray, and red solid lines) but also with the individual shifts induced by the clockwise and counterclockwise orbits (Fig. 3(b) inset).

Adiabatic state evolution is key to sustain particle transport in a topological pump. We investigate the pump adiabaticity by measuring its efficiency for different pump periods, shown in Fig. 4(a). For each value of pump period, we extract the efficiency, defined as the shift in lattice sites per period, by fitting lines to datasets such as Fig. 3. Trivial orbits yield efficiencies around zero for all measured pump

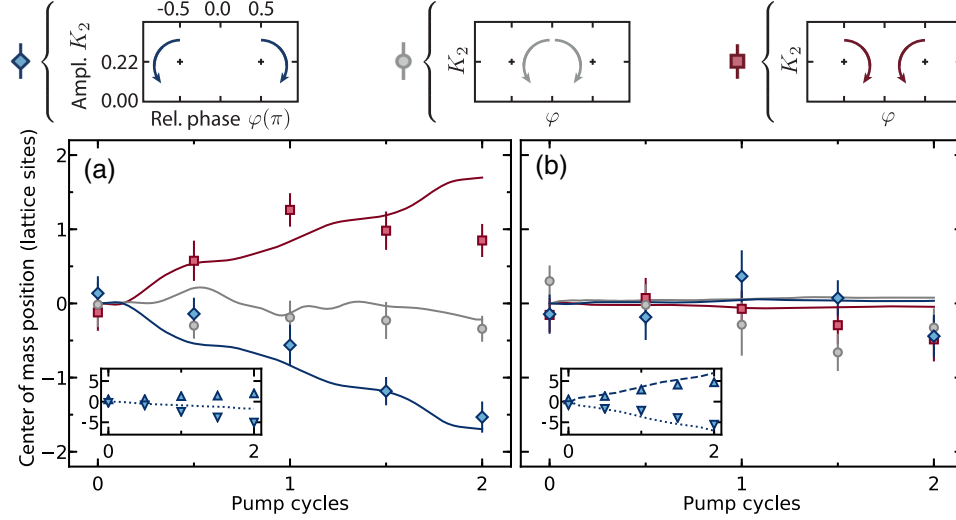


FIG. 3. Topological pumping in a sinusoidal lattice potential. Detected *in situ* center of mass (c.m.) position shown as function of pump cycles. Parameter cycles change the Zak phase by -2π (blue), 0 (gray), and $+2\pi$ (red). Each orbit is a sinusoidal modulation of K_2 and φ centered around $K_2 = 0.20$ and $\varphi = 0, \pm\pi/2$, with amplitudes 0.13 and $\pi/4$, respectively. The other driving parameters are $(\omega/2\pi) = 6.5$ kHz and $K_1 = 0.8$. (a),(b) Averaged c.m. shifts for paired orbits with a pump period of 4 ms (blue diamonds, gray circles, and red squares). The initial state in (a) is the Floquet state prepared with the loading sequence (Fig. 2) while in (b) the pumping waveform is suddenly switched on. Insets show c.m. shifts for clockwise (blue inverted triangles and dotted lines) and counterclockwise (blue upright triangles and dashed lines) blue orbits. The dashed and dotted lines overlap in the inset of panel (a). Each data point of a pumping orbit is the mean and SEM of 12 repetitions. Solid, dashed, and dotted lines are numerical simulations. Theory and experiment differ from quantized pumping due to nonadiabatic effects, while in the experiment deviations from homogeneous band filling can additionally affect the pumping efficiency.

periods (gray circles). In contrast, the efficiency for topological orbits increases from zero for periods below 1 ms to almost unity for periods above 3 ms, saturating at around -0.85 sites per period (blue diamonds).

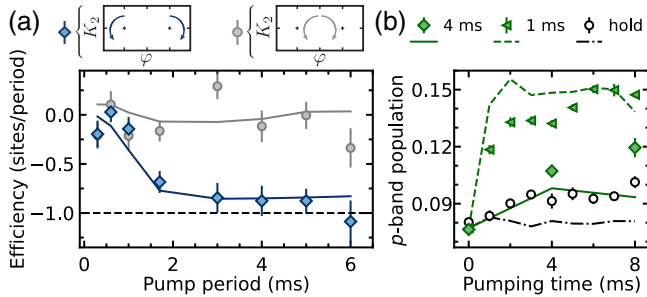


FIG. 4. Adiabatic timescale for topological pumping. (a) Efficiency versus pump period for trivial (gray circles) and topological (blue diamonds) trajectories. Each data point is the mean and error of the fitted slope to a c.m. shift versus pump cycles [datasets such as Fig. 3(a)]. Blue and gray solid lines are numerical simulations. These predict quantized efficiency of -1.0 for periods longer than 12 ms as shown in Fig. S1 in the Supplemental Material [23]. (b) *In-Out* measurement of p -band population versus pumping time for a topological orbit with pump periods 4 ms (green diamonds), 1 ms (green triangles), and hold (black circles). Each data point is the mean and SEM of 3 repetitions. Solid, dashed, and dashed-dotted lines are numerical simulations for the 4 ms, 1 ms, and hold, respectively.

This timescale for adiabatic transport is consistent with the inverse of the minimal gap size $(0.26 \text{ kHz})^{-1} \approx 3.9$ ms, representing an estimate for diabatic transfers to the “excited” Floquet band induced by the pump [36]. Numerical simulations (blue and gray lines) corroborate near-unit efficiency after 3 ms, while quantized efficiency is predicted for periods longer than 12 ms (supplemental material). Residual magnetic and optical potential gradients, inhomogeneous filling, as well as d -band excitations constitute the main limitations for reaching the fully quantized, adiabatic limit (see Supplemental Material [23]).

We perform additional band population measurements to verify the adiabatic timescale for topological transport [Fig. 4(b)]. Specifically, we employ an *In-Out* sequence, similar to the one shown in Fig. 2(f), to detect diabatic transfers due to the pump as accumulated p -band population. Here, the shaking waveform includes loading, pumping, and a time-reversed loading protocol that connects back to the static lattice. We compare the accumulated p -band population between three different cases: pumping with periods of 1 ms (green triangles) and 4 ms (green diamonds), as well as shaking with fixed amplitude K_2 and relative phase φ (black circles, “hold”). The p -band population builds up faster for the 1 ms case than for the 4 ms period, in accordance with measured transport efficiencies of $-0.14(12)$ and $-0.88(15)$ sites per period, respectively. In turn, the 4 ms period shows larger p -band population compared to the hold case, which indicates that

a fully adiabatic pump is expected for longer periods, in agreement with numerical simulations (lines). The small rise of p -band population measured in the no pump (hold) case, which is not predicted by simulations, is attributed to the expansion dynamics of the atomic cloud. Discrepancies in the absolute value of p -band population between theory and experiment can result from an inhomogeneous distribution of quasimomentum states (see Supplemental Material [23]). Overall, the band population measurements support the *in situ* observation of topological transport in the Floquet-engineered Thouless pump.

In conclusion, we demonstrated adiabatic Floquet state preparation and topological pumping of ultracold fermions in a simple sinusoidal potential. The employed loading strategy improves existing methods to prepare topological Floquet states, as the topological gap “slides in” from the BZ edge instead of opening directly at the band inversion points [41]. Moreover, the conceptual simplicity of the driving scheme presented here should be applicable to two-dimensional lattices [33,42–44], establishing a new direction for topological Floquet engineering. Finally, we demonstrated a topological pump that does not rely on superlattice potentials, possibly enabling the metrological application of quantized currents [45] in real materials via two-tone driving [46].

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